

Project Initialization and Planning Phase

Date	13 August 2026
Team ID	-----
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	3 Marks

Project Proposal (Proposed Solution)

Project Overview	
Objective	The main objectives of this project are to study crop production patterns in India, identify areas with high and low crop yields, examine changes in agricultural production over the years, and understand the impact of seasonal variations on crop production across different regions and time periods.
Scope	Analysis of crop production from 1997 to 2021.
Problem Statement	
Description	There is no single visual to show seasonal and yearly trends in crop production.
Impact	This visualization will help policymakers design effective strategies to improve agricultural productivity and make better use of available resources. It will also support agribusiness investors by providing insights into historical crop production, helping them identify profitable crops, emerging opportunities, and regions with steady growth potential.
Proposed Solution	
Approach	Collect the data from 1997 to 2021 on crop production across India from the ministry of agriculture and other agriculture related websites.
Key Features	Agriculture production trends from 1997 to 2021 across India.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	2 x NVIDIA V100 GPUs
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	512GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy
Development Environment	IDE, version control	Jupyter Notebook, Git
Data		
Data	Source	Ministry of Agriculture and Farmers Welfare Directorate of Economics and Statistics